

Systems Analysis and Design

Forest Cover Type Prediction

AUTHORS:

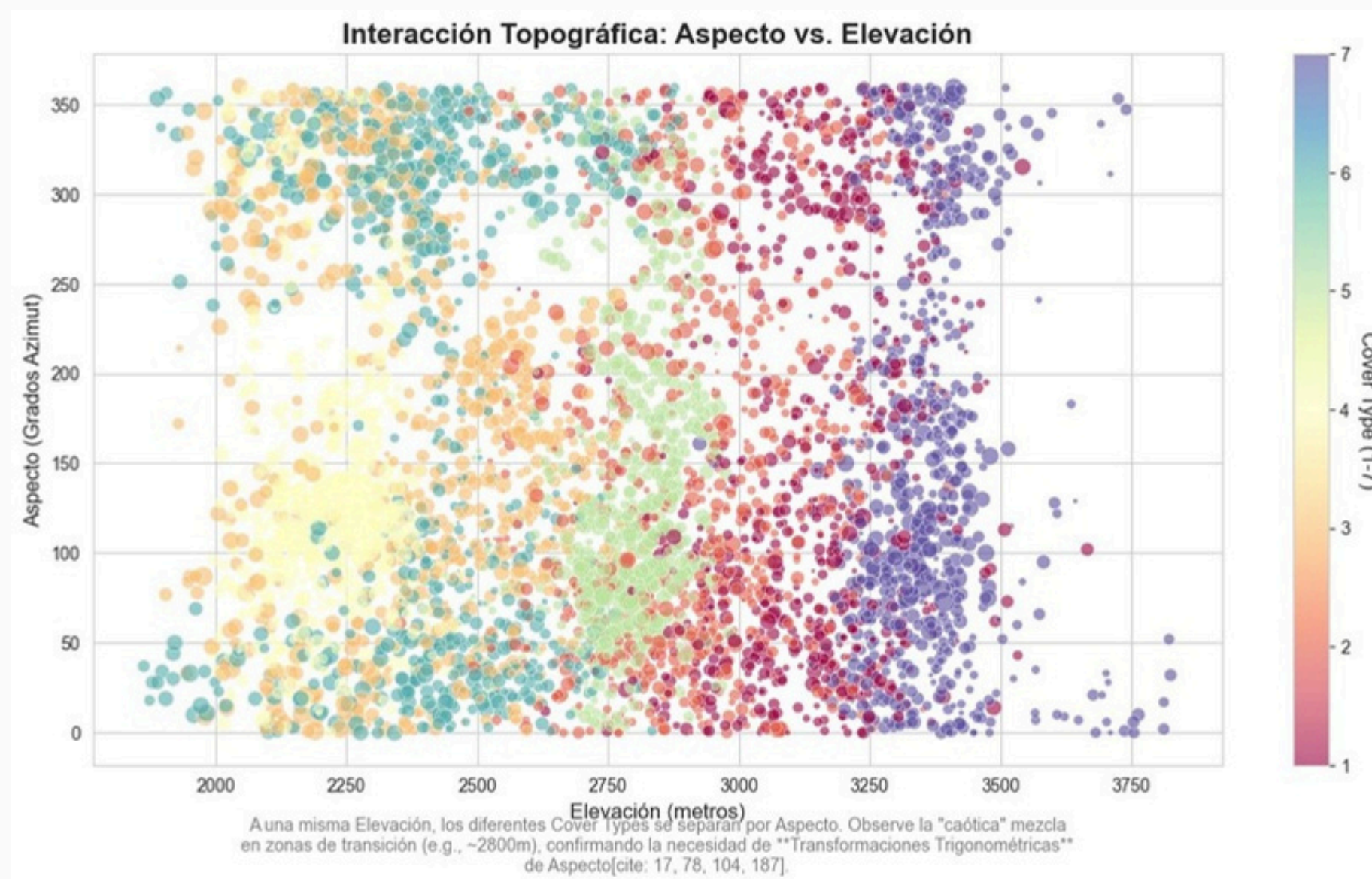
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The Challenge: Ecological Prediction under Complexity

- **Problem:** The Roosevelt National Forest dataset combines 56 features (topographic, soil, and environmental)
 - **Goal:** predict forest cover type among 7 ecological classes.
- **Main challenges:** High dimensionality, nonlinear dependencies, and chaotic elevation-aspect interactions.
 - **Combines:**
 - Machine learning
 - Adaptive systems
 - Sensitivity, chaos, and simulations.

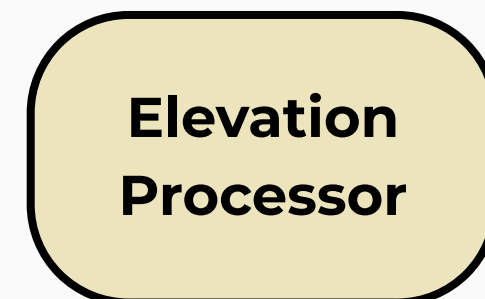
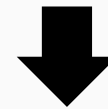
System Vulnerabilities and Chaos



- Elevation thresholds (2400m, 2800m, 3200m) act as ecological tipping points.
- Small data changes can completely alter predictions (butterfly effect).
- Soil-type sparsity and limited geography create risk of overfitting.

Preparation of Data

3 Layers for data preparation



- Classifies elevation into ecological zones.
 - Detects chaotic areas near critical thresholds (± 50 m).
 - Calculates distance to the threshold and applies an uncertainty factor of x2.
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- Corrects angle circularity.
 - Converts aspect (0-360°) to sine and cosine components.
 - Fully preserves the original direction.
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- Reduces 40 highly dispersed soil types to 15 groups.
 - Decreases sparsity and improves model stability.
 - Stored in **train_transformed**.
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LightGBM

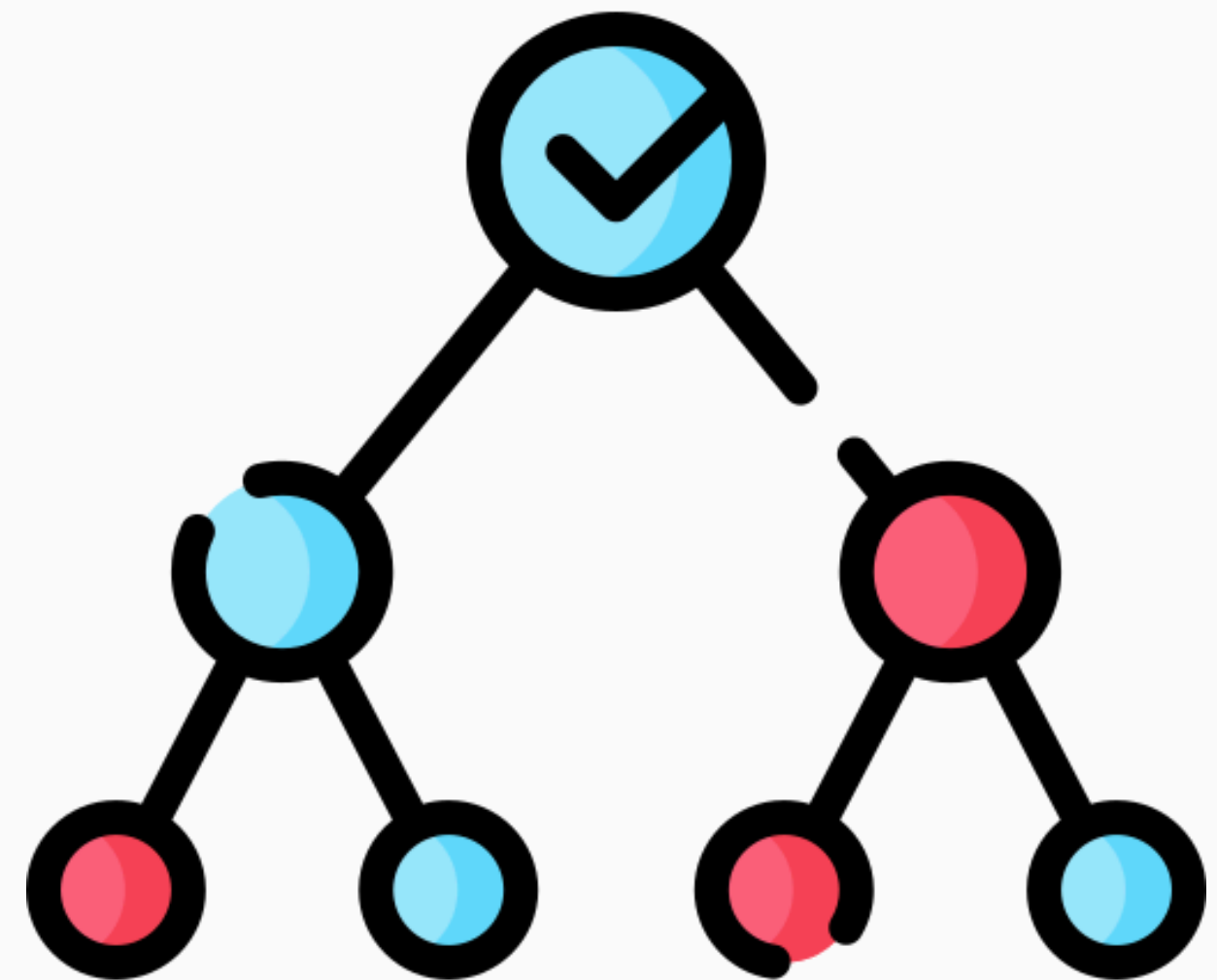
The Selected Model

It is a tree-based boosting algorithm with histogram learning

- Ideal for: Tabular data with mixed features (numeric + categorical)
- Configuration: Hyperparameters optimized to avoid overfitting, maximize speed, and accuracy

Key Metrics:

- Validation Accuracy: 87.55%
- CV Accuracy (5-fold): 94.60% $\pm$ 1.20%
- Training Time: 7.4 seconds
- Inference Latency: <50ms
- Model Size: 2.3 MB



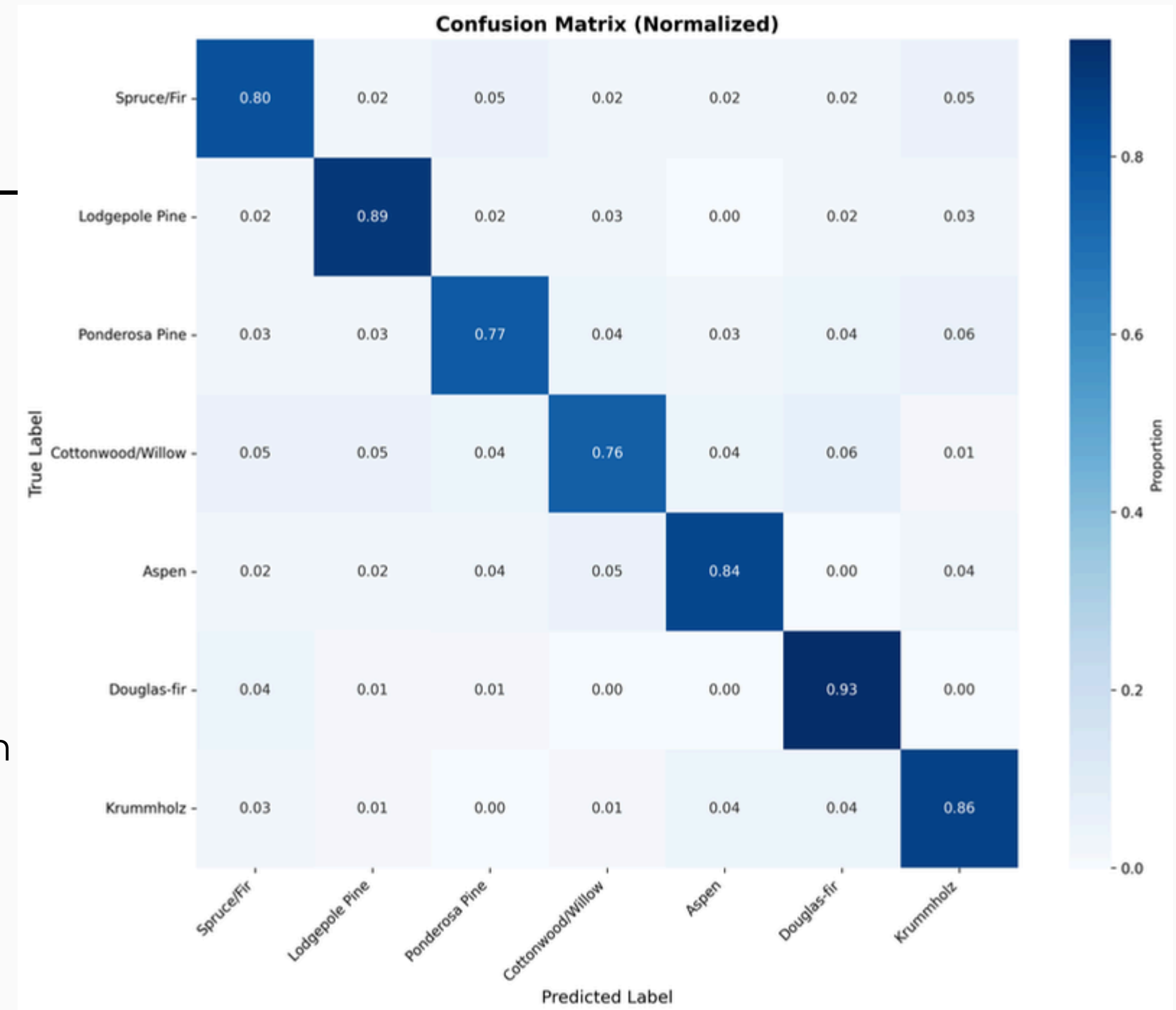
LightGBM Result Analysis

Best Performance:

- Douglas fir: 93% accuracy
- Lodgepole Pine: 89%
- Krummholz: 86%

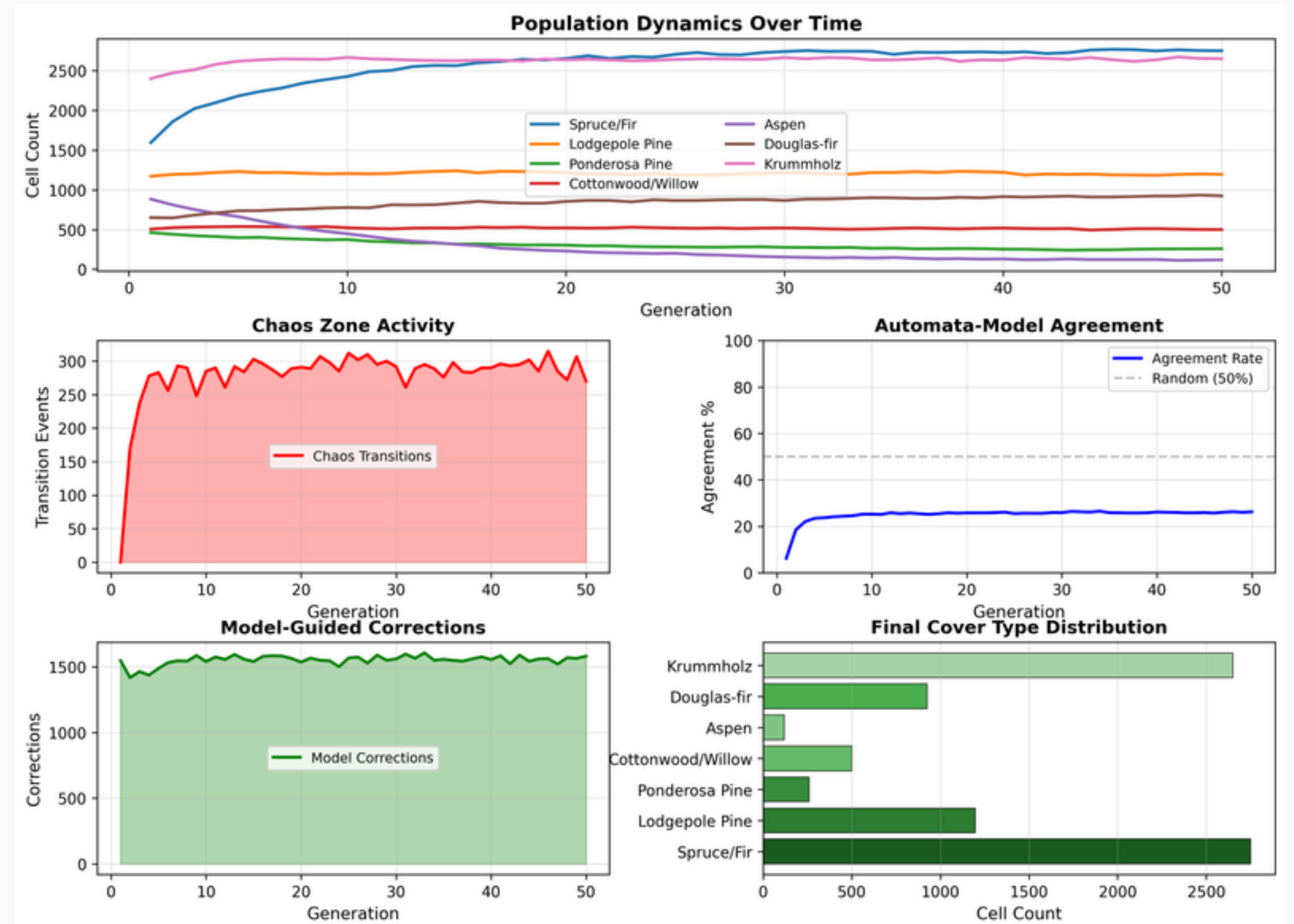
Highest Confusion:

- Ponderosa Pine: 77% (confused with Lodgepole 3%, Aspen 3%)
- Cottonwood/Willow: 76% (confused with Douglas fir 6%, Aspen 4%)

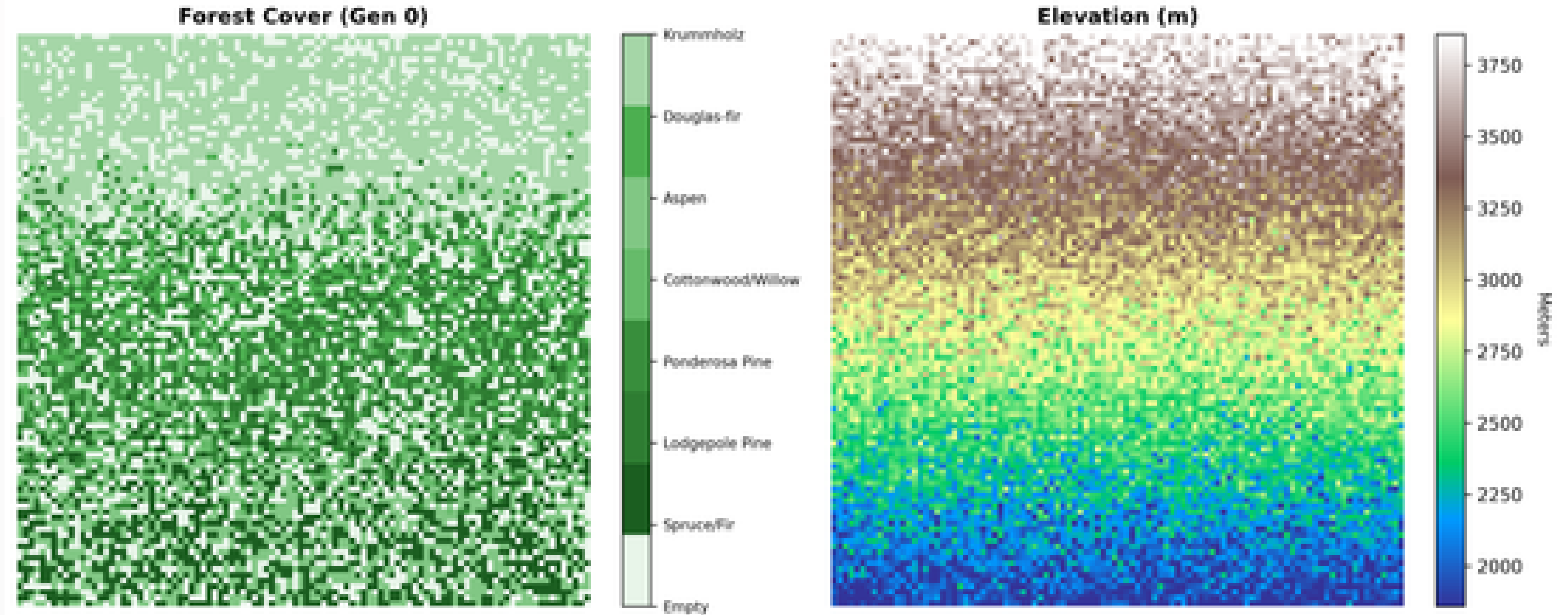


Simulations 1°

- Training with LightGBM / Random Forest / XGBoost.
- Five-step cross-validation.
- Optimization with Optuna (100 trials).
- Metrics analyzed: Precision, F1-macro, LogLoss.
- Monitoring of model uncertainty and sensitivity.

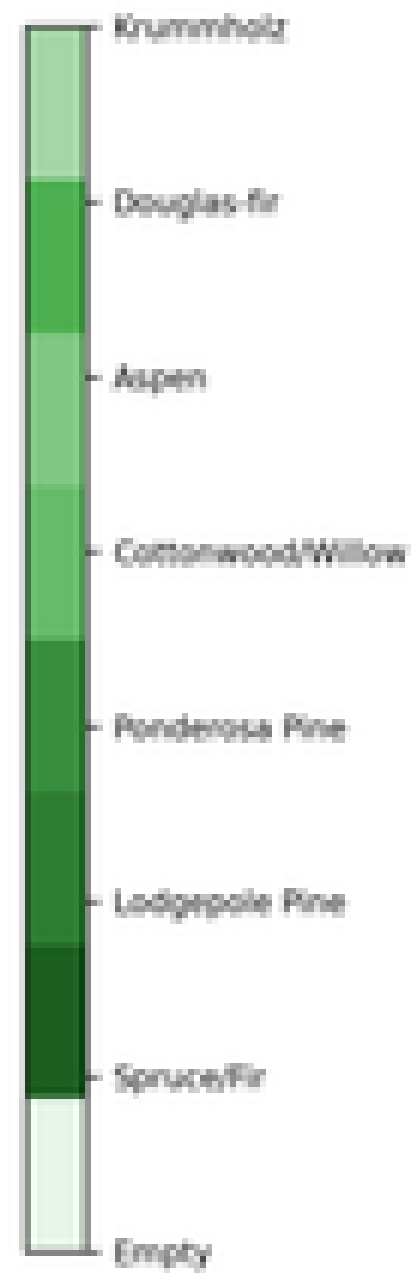
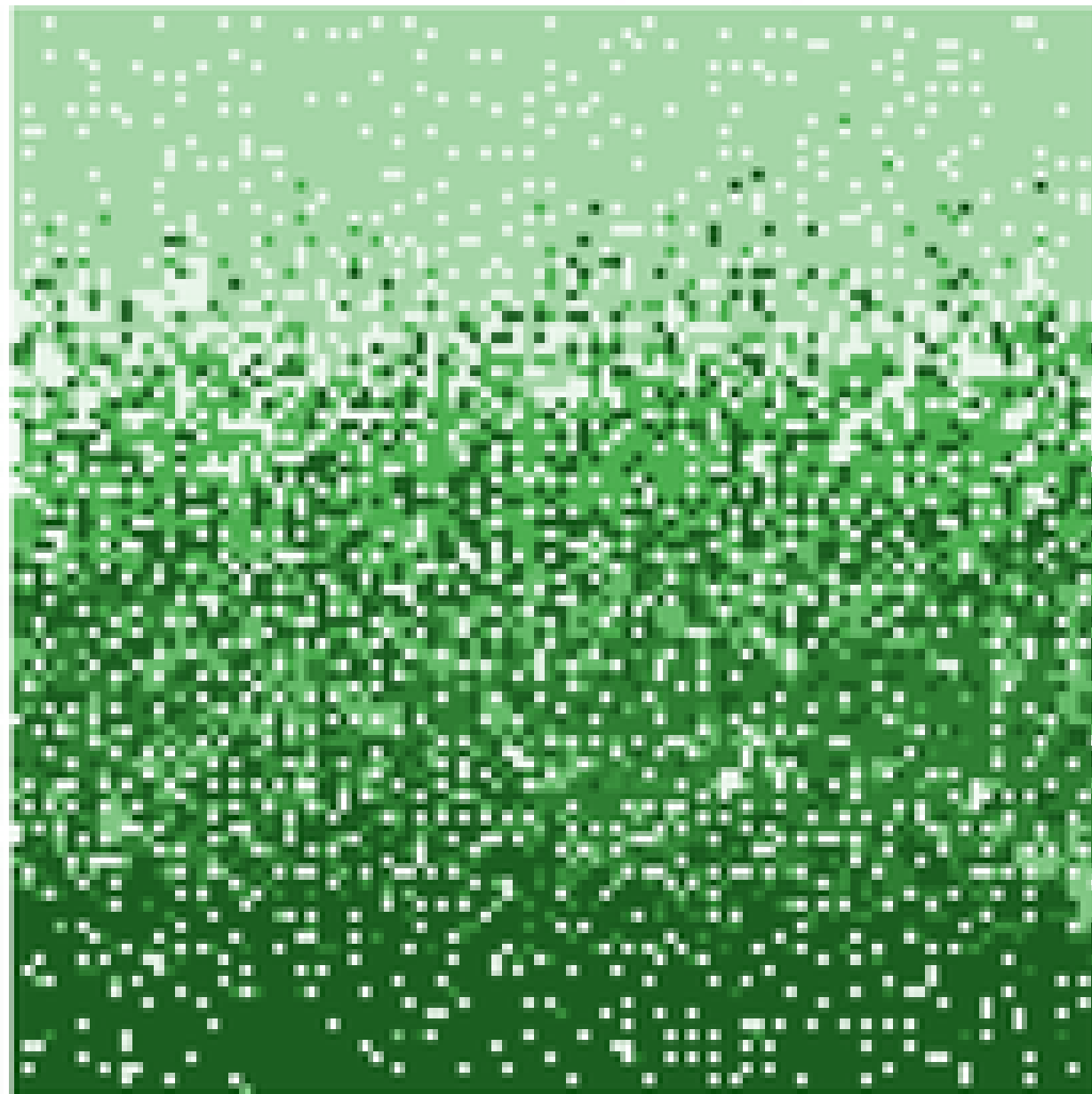


Simulations 2°

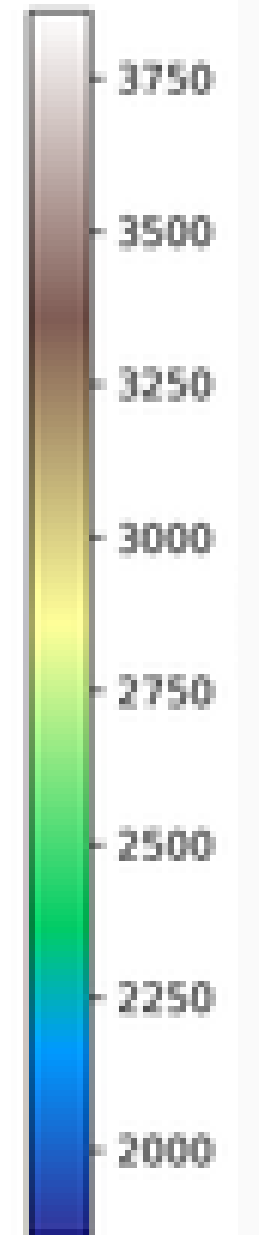
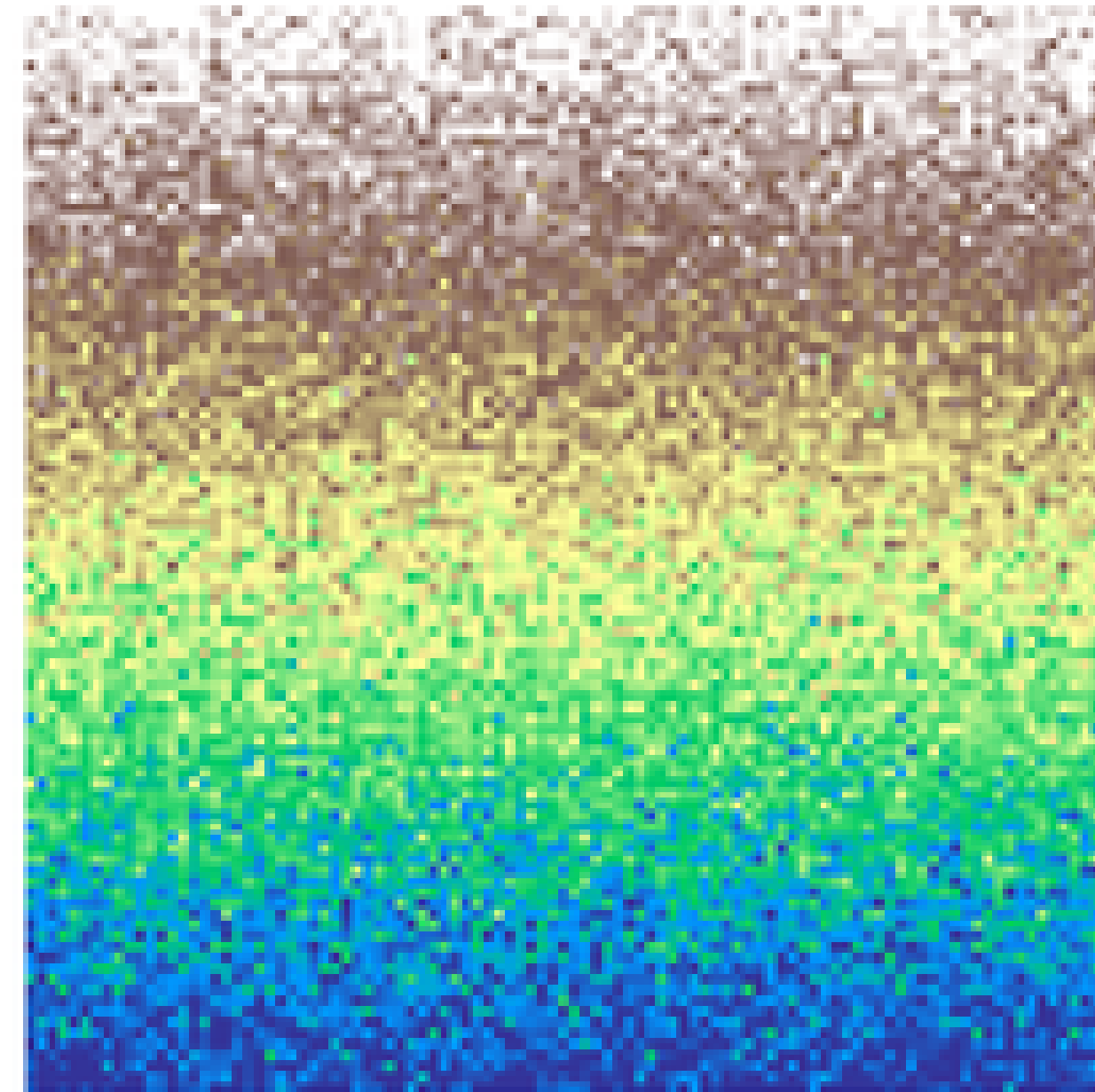


Simulations 2°

Forest Cover (Gen 50)



Elevation (m)



Final Conclusions

Key Achievements

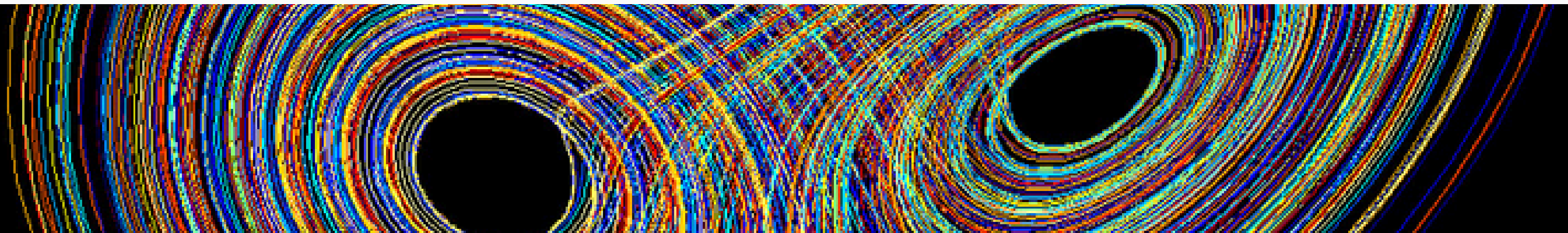


- Predictive ML system with a 4-layer architecture for forest cover
- Chaos-aware uncertainty quantification
- Hybrid validation (cellular automata + ML)
- Full MLOps pipeline in production

Research Questions



- Chaos improves reliability in ecological systems
- Feature engineering improves interpretability
- Optimal LightGBM (balance of accuracy and speed)
- Hybrid simulations validate beyond traditional metrics





Thank you for your time.

Several parts of the project are summarized here, but if you want to analyze our system in great detail, here is the GitHub.